SQE Assignment 2: Testing Report

Student Course Registration System (registration.py), SE3002 Software Quality Engineering

Team: [member names and roll numbers]. Repository: [GitHub URL]. Run date: 4 November 2026.

1. Summary

Black-box tests were designed from REQUIREMENTS.md before reading the code. Running them on the unmodified source gave 79 executions, 58 passed and 21 failed. The failures trace to five distinct defects (DEF-01 to DEF-05, section 7). White-box analysis of registration_decision() gave V(G) = 10 (condition-level) and exposed that the WAITLISTED branch is unreachable. Coverage rose from 93% to 95% with the white-box tests; the remaining gap (lines 59 and 71) is caused by DEF-04 and closes to 100% once the defects are fixed. With the five fixes applied in a scratch copy, all 97 tests pass.

2. Black-box design (Part A)

2.1 Equivalence class partitioning (A1)

27 ECP cases cover credit hours (valid, exceeds, invalid current, invalid course, validation order), dues (positive, zero, negative), prerequisites (none required, all, some, none completed, extra), already-registered status, availability (seats free, full) and invalid capacity or enrollment. Each row in docs/blackbox_test_design.xlsx (sheet A1 ECP) records requirement ID, class, representative input, expected output and test ID.

2.2 Boundary value analysis (A2)

33 BVA cases: current credits -1, 0, 18, 19; course credits 0, 1, 4, 5; combined totals 18 and 19 (14+4, 15+4, 15+3, 16+3, 17+1, 17+2, 18+1); enrolled = capacity-1, capacity, capacity+1 and -1; capacity 0 and 1; dues -1, 0, 1; prerequisite sets one short of complete. See sheet A2 BVA.

2.3 Decision table (A3)

Conditions: D dues > 0, A already registered, P prerequisite missing, L credit limit exceeded, S course full. Priority D > A > P > L > S (FR-09). A dash means the condition is not evaluated because a higher-priority rule already decided.

Rule	D	A	P	L	S	Outcome	Test
R1	T	-	-	-	-	DUES_BLOCK	DT-01
R2	F	T	-	-	-	ALREADY_REGISTERED	DT-02
R3	F	F	T	-	-	PREREQUISITES_MISSING	DT-03
R4	F	F	F	T	-	CREDIT_LIMIT	DT-04
R5	F	F	F	F	T	WAITLISTED	DT-05
R6	F	F	F	F	F	REGISTERED	DT-06

Priority combinations DT-07 to DT-15: D+A+P+L+S, A+P+L+S, P+L+S, L+S, D+S, A+P, D+A, A+L, P+L, each with the higher-priority outcome expected. 6 outcomes, 15 decision tests.

2.4 Pytest implementation (A4)

tests/test_blackbox.py holds the technique cases (parametrized, with pytest.raises(ValueError, match=...) for error cases) and 21 state-change tests (ST-xx): registration success, waitlist append, no duplicate waitlist entries, every unsuccessful outcome leaving both objects equal to a deep-copied snapshot, registration_decision not mutating (FR-10), invalid input leaving state unchanged, withdrawal success and failure, register-then-withdraw round trip. Expected values come from the SRS only. Initial output is preserved in docs/blackbox_initial_run.txt.

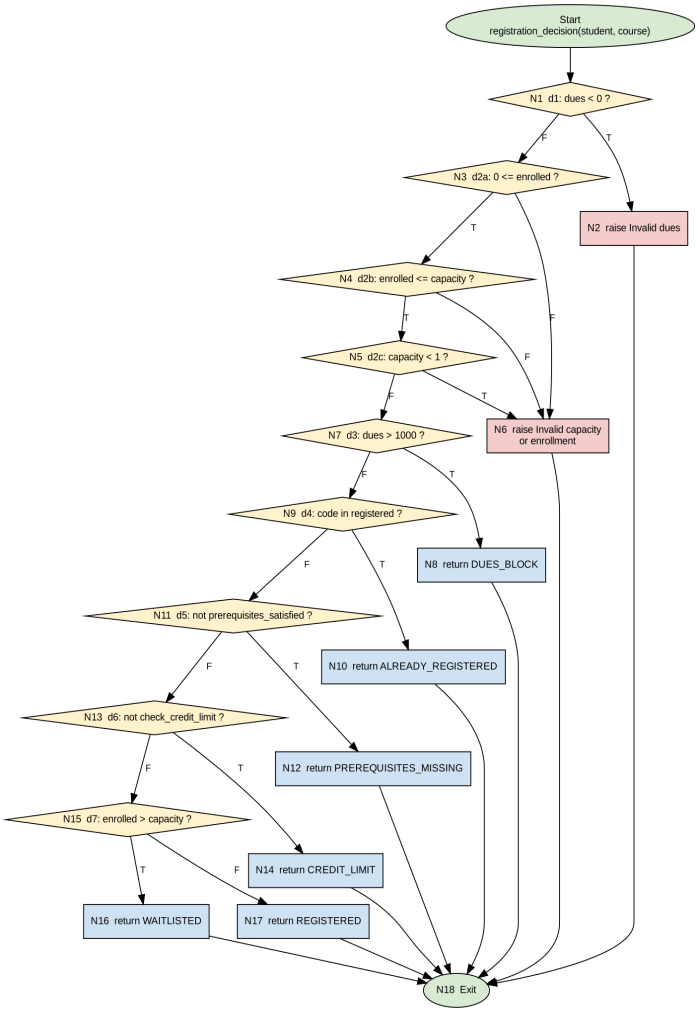
2.5 Optimization (A5)

75 technique cases were designed (27 ECP + 33 BVA + 15 DT). A case is an exact duplicate only if the function, the fully-resolved inputs (defaults filled in) and the expected result all match. Identical outputs alone did not qualify: for example dues = 1 (BVA-21) and dues = 500 (ECP-10) both give DUES_BLOCK but test different inputs, so both are kept. 17 duplicates were merged, leaving 58 unique executions. Each merged test keeps a combined pytest ID so traceability is retained.

Executed as	Merged technique IDs
ECP-11+ECP-14+BVA-20+DT-06	dues zero, no prerequisites, dues = 0, all conditions false
ECP-19+BVA-24+DT-05	course full (30/30)
ECP-23+BVA-31+DT-04	credits 16 + 3 = 19
ECP-10+DT-01	dues 500
ECP-13+DT-02	already registered
ECP-17+DT-03	prerequisite missing
ECP-20+BVA-29	capacity 0
ECP-21+BVA-25	enrolled 31 of 30
ECP-22+BVA-26	enrolled -1
ECP-02+BVA-11	check_credit_limit(16, 3)
ECP-05+BVA-05	check_credit_limit(5, 0)
ECP-31+BVA-42	prerequisites all completed
ECP-32+BVA-41	prerequisites some completed

3. White-box analysis (Part B)

3.1 Control flow graph and cyclomatic complexity (B1)



Convention: the Start oval is an entry marker and is not counted; there is one exit node (N18); every raise or return node connects to it. The compound condition *not 0 <= enrolled <= capacity or capacity < 1* is expanded into three atomic predicate nodes (N3, N4, N5) because Python short-circuits; calls to prerequisites_satisfied and check_credit_limit are single predicates (N11, N13).

Method	Value
Condition-level: E = 26, N = 18	V(G) = E - N + 2 = 10
Condition-level: predicate nodes = 9 (N1, N3, N4, N5, N7, N9, N11, N13, N15)	predicates + 1 = 10 (matches)
Decision-level (compound as one node): E = 22, N = 16	V(G) = 22 - 16 + 2 = 8; predicates 7 + 1 = 8 (matches)

The reported value is 10, the condition-level figure, because it counts each short-circuit branch as an independent route.

Independent basis path set (10 paths)

Path	Nodes	Input	Expected	Test
P1	N1-N2-N18	dues -1	ValueError Invalid dues	WB-P01
P2	N1-N3-N6-N18	enrolled -1	ValueError capacity/enrollment	WB-P02
P3	N1-N3-N4-N6-N18	enrolled 31, capacity 30	ValueError	WB-P03
P4	N1-N3-N4-N5-N6-N18	capacity 0, enrolled 0	ValueError	WB-P04
P5	...N5-N7-N8-N18	dues 1001	DUES_BLOCK	WB-P05

Path	Nodes	Input	Expected	Test
P6	...N7-N9-N10-N18	already registered	ALREADY_REGISTERED	WB-P06
P7	...N9-N11-N12-N18	prerequisite missing	PREREQUISITES_MISSING	WB-P07
P8	...N11-N13-N14-N18	16 + 3 credits	CREDIT_LIMIT	WB-P08
P9	...N13-N15-N16-N18	enrolled 30/30	WAITLISTED	WB-P09 (fails: DEF-04, path infeasible in current code)
P10	...N15-N17-N18	defaults	REGISTERED	WB-P10

N15 (enrolled > capacity) can never be true after the validation at N3 to N5, so P9 and the WAITLISTED return are dead code. This is the structural view of DEF-04.

3.2 Statement and branch coverage (B2)

Stage	Tests	Stmts	Miss	Branch	BrPart	Cover	Uncovered
Initial: black-box only	79 (21 fail)	57	3	26	3	93%	51, 59, 71
After white-box additions	97 (25 fail)	57	2	26	2	95%	59, 71
After defect fixes (scratch copy)	97 (0 fail)	58	0	26	0	100%	none

Gaps found: line 51 (DUES_BLOCK return) was never reached because black-box dues values were below 1000, which itself exposes DEF-03. WB-P05 (dues 1001) covers it. Lines 59 (WAITLISTED return) and 71 (waitlist append) are unreachable until DEF-04 is fixed. Added tests: WB-P01 to WB-P10, WB-MCDC-T1 to T4, WB-G01 to G04.

3.3 Condition coverage and MC/DC (B3)

Selected decision (the only compound one): raise if **(not (A and B)) or C** where A: 0 <= enrolled, B: enrolled <= capacity, C: capacity < 1. NE means not evaluated because of short-circuiting.

Test	A	B	C	Decision (raise?)	Inputs	pytest
T1	F	NE	NE	T	enrolled -1, capacity 30	WB-MCDC-T1
T2	T	F	NE	T	enrolled 31, capacity 30	WB-MCDC-T2
T3	T	T	T	T	enrolled 0, capacity 0	WB-MCDC-T3
T4	T	T	F	F	enrolled 0, capacity 30	WB-MCDC-T4

Condition	Pair	Effect shown
A	T1, T4	A changes F to T while the outcome changes T to F; B and C are masked in T1 (not evaluated)
B	T2, T4	A = T in both; B changes F to T and the outcome changes T to F; C is masked in T2
C	T3, T4	A = B = T in both; C changes T to F and the outcome changes T to F

Four tests (N + 1 for N = 3) achieve masking MC/DC. The combinations A = F with B or C evaluated cannot occur, so condition coverage has 4 reachable rows, not 8. Computed by hand; pytest-cov does not measure this.

3.4 White-box pytest tests (B4)

tests/test_whitebox.py has 18 tests; the docstring of each names the path, branch or condition it targets. WB-G01 targets the register_course waitlist-append True branch (line 71); WB-G02 the already-on-waitlist False side of the 'and'; WB-G03 the neither-REGISTERED-nor-WAITLISTED path; WB-G04 both branches of withdraw_course.

4. Defects (Part C3)

ID	Requirement	Expected	Actual	Failed tests	Issue / status
DEF-01	FR-01, FR-07	check_credit_limit(15, 3) = True	False (uses < 18)	BVA-08, BVA-10, BVA-12, BVA-30, BVA-34	[#1] Open until fix PR merges
DEF-02	FR-03, FR-06	some prerequisites missing gives False	True (uses any, not all)	ECP-32+BVA-41, ECP-16	[#2] Open
DEF-03	FR-04	dues Rs. 1 gives DUES_BLOCK	REGISTERED (threshold 1000)	ECP-10+DT-01, BVA-21, DT-07, DT-11, DT-13, ST05 x2	[#3] Open
DEF-04	FR-08, FR-12	full course gives WAITLISTED	REGISTERED (uses >, dead branch)	ECP-19+BVA-24+DT-05, BVA-28, ST03, ST04	[#4] Open
DEF-05	FR-13	withdraw decrements enrolled	enrolled unchanged	ST11, ST13, ST14	[#5] Open

Fixes (one line each, in defect-fixes.patch): change < to <=; use required.issubset(completed); change 1000 to 0; change > to >= for capacity; add course.enrolled -= 1 in withdraw_course. With them applied all 97 tests pass and coverage is 100%. Replace #n above with the real issue numbers after creating the issues from docs/ISSUES.md.

5. GitHub evidence (Parts C1 to C5)

Item	Link or note
Repository	[URL]
Branches	main, blackbox-testing, whitebox-testing, defect-fixes
Issues	[#1 to #5 links]
Pull requests	[PR-1 blackbox-testing, PR-2 whitebox-testing, PR-3 defect-fixes; reviewer for each]
Actions runs	[failing run before fixes link], [passing run after fixes link]

6. Member contributions

Member	Contribution	Commits / PRs reviewed
[name]	[e.g. ECP/BVA design, test_blackbox.py]	[ ]
[name]	[e.g. CFG, MC/DC, test_whitebox.py]	[ ]
[name]	[e.g. defect fixes, Actions workflow]	[ ]

7. Reproduction

python -m pytest tests/ --cov=registration --cov-branch --cov-report=term-missing. Raw outputs: docs/blackbox_initial_run.txt, docs/all_tests_before_fix.txt, docs/coverage_initial.txt, docs/coverage_final_before_fix.txt, docs/coverage_after_fix.txt.